Gravitational Memory Workbook

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Code used while writing astrophysics senior thesis at Haverford College entitled “The Effect of Gravitational Memory on CMB Light.”

Defining Variables and Functions

In[ ]:=
```
ClearAll["Global`*"]
```

Rotation matrices and GW polarization tensors for a GW traveling along the z-axis.

In[ ]:=
```
Rz[ϕ_] := {{Cos[ϕ], -Sin[ϕ], 0}, {Sin[ϕ], Cos[ϕ], 0}, {0, 0, 1}} (*z-axis rotation*)
```

In[ ]:=
```
P = {{1, 0, 0}, {0, -1, 0}, {0, 0, 0}}; (*+ polarized wave*)
G = {{p, x, 0}, {x, -p, 0}, {0, 0, 0}}; (*general wave polarization*)
X = {{0, 1, 0}, {1, 0, 0}, {0, 0, 0}}; (*cross polarized wave*)
```

Applying the rotation matrix to + polarization. Tensors rotate as $T' = RTR^{-1}$

To get the new tensor you apply the rotation matrix R and its inverse R^{-1}

In[ ]:=
```
Simplify[Rz[ϕ].P.Transpose[Rz[ϕ]]] // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \text{Cos}[2\phi] & \text{Sin}[2\phi] & 0 \\ \text{Sin}[2\phi] & -\text{Cos}[2\phi] & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Rotations about the y and z-axes

In[ ]:=
```
Ry[α_] := {{Cos[α], 0, Sin[α]}, {0, 1, 0}, {-Sin[α], 0, Cos[α]}}
(*y-axis rotation*)
Rx[α_] := {{1, 0, 0}, {0, Cos[α], -Sin[α]}, {0, Sin[α], Cos[α]}}
 (*x-axis rotation*)
```

Variables and Functions

Constants:

c=1,

dG Earth to GW source distance,

tP time photon is received on Earth,
tG time GW is received on Earth which we set to 0,
(ϕ,θ) location on Earth's sky where the photon is received
h0 GW strain amplitude measured on Earth
t time as measured on Earth

Functions:
Functions are capitalized. A is the function that gives α for given parameters, RP is the function that gives rP.
rP is the distance from GW source to photon
α is the angle between GW source and photon (measured up from GW propegation vector)
β is the azimuthal angle from the GW source to the photon
dP is the distance from Earth to the photon
tmin is the initial time the photon encounters the GW
n is the vector pointing to point on the sky where the photon is found

In[∘]:=
```
$Assumptions = dG > 0 && h0 > 0 && tP > tG && 0 ≤ θ ≤ 2 π && t ∈ Reals;
c = 1;
tG = 0;
M[dP_, dG_] := If[dP > dG, ArcCos[dG/dP], 0];
(*A[dP_, dG_, θ_] :=  If[θ≤M[dP, dG],ArcTan[(dP Sin[θ])/(dG - dP Cos[θ])]+π,
  If[θ≤2π-M[dP, dG],ArcTan[(dP Sin[θ])/(dG - dP Cos[θ])],ArcTan[(dP Sin[θ])/(dG - dP Cos[θ])]-π]] *)
DP[tP_, t_] := c (tP - t)
(*DP[tP_, t_] := If[tP>t,c(tP-t), 0]*)
(*solves for if tP > t, but shouldn't be needed,
although it did change the ϵ and h expresions a little*)
Tmin[tG_, tP_, dG_, θ_] := (-c tG^2 + 2 dG tG + c tP^2 - 2 dG tP Cos[θ])/(-2 c tG + 2 dG + 2 c tP - 2 dG Cos[θ])
RP[dG_, tP_, t_, θ_] := Sqrt[dG^2 + (tP - t)^2 - 2 dG (tP - t) Cos[θ] ]
n[θ_] := {Sin[θ], 0, Cos[θ]} (*2D looking vectore in x/z plane*)
v[θ_, ϕ_] := {Sin[θ] Cos[ϕ], Sin[θ] Sin[ϕ], Cos[θ]} (*3D looking vector *)
```

Functions for α and β, which are related to polar angles in the GW source's coordinate system. These are the angles we rotate through to relate the polarization metric seen at Earth to the polarization metric seen by a photon at a given time.

In[ ]:=
```
A[dP_, dG_, θ_, ϕ_] := If[θ ≤ M[dP, dG], ArcTan[(dP Sin[θ] Cos[ϕ])/(dG - dP Cos[θ])] + π,
   If[θ ≤ 2 π - M[dP, dG], ArcTan[(dP Sin[θ] Cos[ϕ])/(dG - dP Cos[θ])], ArcTan[(dP Sin[θ] Cos[ϕ])/(dG - dP Cos[θ])] - π]]
B[dP_, dG_, θ_, ϕ_] := If[θ ≤ M[dP, dG], ArcTan[(dP Sin[θ] Sin[ϕ])/(dG - dP Cos[θ])] + π,
   If[θ ≤ 2 π - M[dP, dG], ArcTan[(dP Sin[θ] Sin[ϕ])/(dG - dP Cos[θ])], ArcTan[(dP Sin[θ] Sin[ϕ])/(dG - dP Cos[θ])] - π]]
```

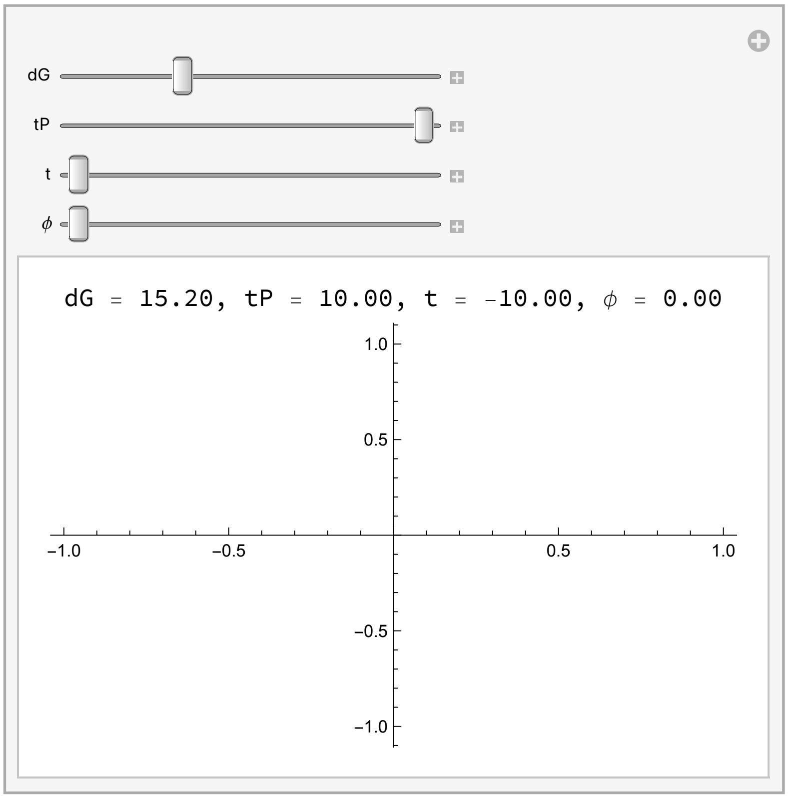
The trig expressions Sin[2 ArcTan [*x*]] , Cos[2 ArcTan [*x*]], and Cos[4 ArcTan[x]] appears frequently in our code. Here, we define functions which can be used to simplify these expressions automatically.

In[ ]:=
```
SinSimp[opp_, adj_] := FullSimplify[2 * opp/Sqrt[adj^2 + opp^2] * adj/Sqrt[adj^2 + opp^2]]
(*Simplifies Sin[2ArcTan[opp/adj]]*)(*Sin(2x) = 2CosxSinx*)
CosSimp[opp_, adj_] := FullSimplify[(adj/Sqrt[adj^2 + opp^2])^2 - (opp/Sqrt[adj^2 + opp^2])^2]
(*Simplifies Cos[2ArcTan[opp/adj]]*)(*Cos(2x) = Cos^2(x) - Sin^2(x)*)
CosSimp4AT[opp_, adj_] :=
 8 (adj/Sqrt[adj^2 + opp^2])^4 - 8 (adj/Sqrt[adj^2 + opp^2])^2 + 1 (*Cos[4x] = 8Cos[x]^2-8Cos[x]^2+1)*)
```

Testing functions

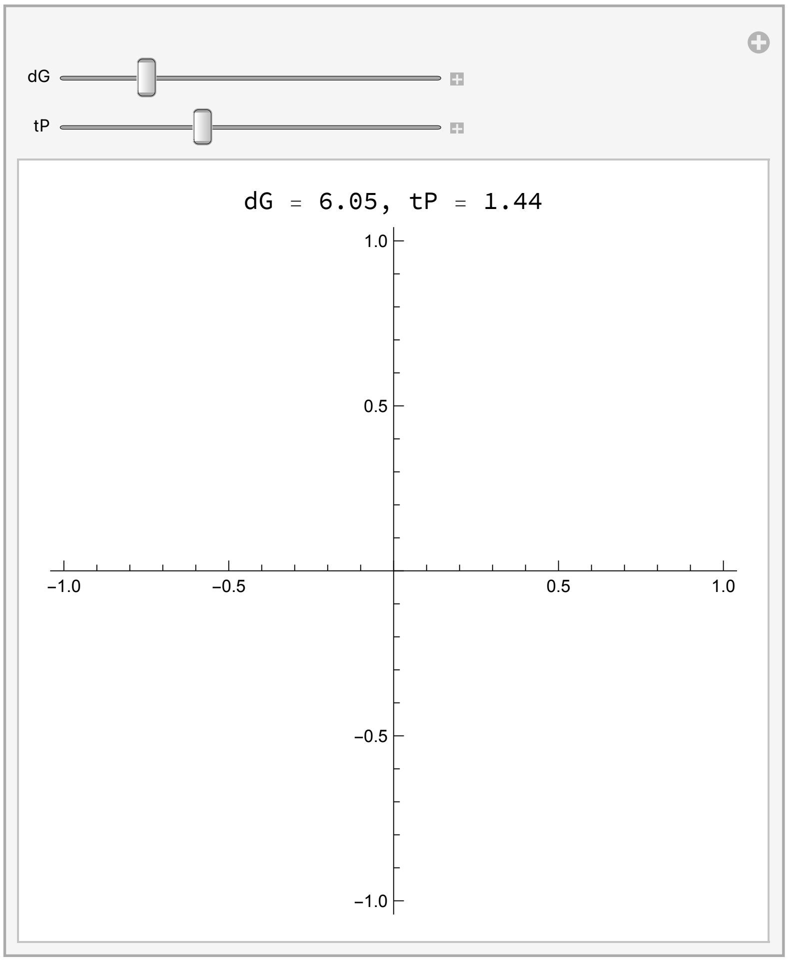
```
Manipulate[Labeled[Plot[A[DP[tP, t] , dG, θ, ϕ], {θ, 0, 2 π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
    ", t = ", NumberForm[t, {3, 2}], ", ϕ = ", NumberForm[ϕ, {3, 2}]}], Top],
 {dG, 0.1, 50}, {tP, 0.1, 10}, {t, -10, -0.1}, {ϕ, 0, 2 π}]
(*Plot of the angle α. This contains If statments so it's
 important to make sure the final plot is continuous.*)
```

Out[ ]=



```
Manipulate[Labeled[PolarPlot[tP - Tmin[tG, tP, dG, θ] , {θ, 0, 2 π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}]}], Top],
 {dG, 0.1, 30}, {tP, 0.01, 4}] (*This is a polar plot of the total amout of time
  a photon has spent in the perturbed metric as a function of the distance to the
  GW and the time the photon in recived on Earth. The time the GW passed is t=
 0 and the GW source is along the z-axis (θ=0).*)
```

Out[∘]=



Isotropic, Plus Polarized Metric in 2D Simplification

Isotropic, Plus Polarized Metric in 2D binary’s plane: Work in Progress

3D, Isotropic, Plus-Polarised Wave: Work in Progress

```
h3DIso =
  Simplify[Rx[B[DP[tP, t], dG, θ, ϕ]].Ry[-A[DP[tP, t], dG, θ, ϕ]].P/RP[dG, tP, t, θ].
    Transpose[Ry[-A[DP[tP, t], dG, θ, ϕ]]].Transpose[Rx[B[DP[tP, t], dG, θ, ϕ]]]];
MatrixForm[h3DIso]
```

Out[ ]//MatrixForm=

$$
\begin{pmatrix}
\dfrac{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]}\;\left(dG^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]+(t-tP)^2\,\mathrm{Cos}[\theta]^2+(t-tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)} \\[2em]
\begin{cases}
-\left(\left((t-tP)\,\mathrm{Sin}[\theta]\,\mathrm{Sin}[\phi]\,\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-t+tP)\,\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]}{dG+(t-tP)\,\mathrm{Cos}[\theta]}\right]\right]\right)\Big/\left(2\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)\right)}\right)\right) & \left(dG+t\geq tP\;||\;\theta\leq\mathrm{ArcCos}\left[\frac{dG}{-t+tP}\right]\;||\;\theta+\mathrm{ArcCos}\left[\frac{dG}{-t+tP}\right]>2\,\pi\right)\;\&\&\;(dG+t<tP\;||\;\theta\leq 0) \\
\left((t-tP)\,\mathrm{Sin}[\theta]\,\mathrm{Sin}[\phi]\,\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-t+tP)\,\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]}{dG+(t-tP)\,\mathrm{Cos}[\theta]}\right]\right]\right)\Big/\left(2\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)\right)}\right) & \mathrm{True}
\end{cases} \\[2em]
\begin{cases}
-\left(\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-t+tP)\,\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]}{dG+(t-tP)\,\mathrm{Cos}[\theta]}\right]\right]\Big/\left(2\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)\right)}\right)\right) & \left(dG+t\geq tP\;||\;\theta\leq\mathrm{ArcCos}\left[\frac{dG}{-t+tP}\right]\;||\;\theta+\mathrm{ArcCos}\left[\frac{dG}{-t+tP}\right]>2\,\pi\right)\;\&\&\;(dG+t<tP\;||\;\theta\leq 0) \\
\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-t+tP)\,\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]}{dG+(t-tP)\,\mathrm{Cos}[\theta]}\right]\right]\Big/\left(2\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)\right)}\right) & \mathrm{True}
\end{cases}
\end{pmatrix}
$$

Applying Trig Simplifications

In[ ]:=
```
SinSimp[opp, adj] (*Simplifies Sin[2ArcTan[opp/adj]]*)
```

Out[ ]=

$$\frac{2\ \text{adj opp}}{\text{adj}^2+\text{opp}^2}$$

In[ ]:=
```
h3DIso = FullSimplify[
```

$$\left(\frac{(dG+(t-tP)\ Cos[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]}\ \left(dG^2+2\ dG\ (t-tP)\ Cos[\theta]+(t-tP)^2\ Cos[\theta]^2+(t-tP)^2\ Cos[\phi]^2\ Sin[\theta]^2\right)}\right.$$

$$\begin{cases} -\left(((t-tP)\ Sin[\theta]\ Sin[\phi]\ SinSimp[(-t+tP)\ Cos[\phi]\ Sin[\theta],\ dG+(t-tP)\ Cos[\theta]])\middle/\left(2\ (dG+(t-tP)\ Cos[\theta])\sqrt{\left(\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]\right)\left(1+\frac{(t-tP)^2\ Sin[\theta]^2\ Sin[\phi]^2}{(dG+(t-tP)\ Cos[\theta])^2}\right)\right)}\right)\right) & \left(dG+t\geq tP\ ||\ \theta\leq ArcCos\left[\frac{dG}{-t+tP}\right]\ ||\ \theta+ArcCos\left[\frac{dG}{-t+tP}\right]>2\pi\right)\ \&\&\ (dG+t<tP\ ||\ \theta\leq 0) \\ ((t-tP)\ Sin[\theta]\ Sin[\phi]\ SinSimp[(-t+tP)\ Cos[\phi]\ Sin[\theta],\ dG+(t-tP)\ Cos[\theta]])\middle/\left(2\ (dG+(t-tP)\ Cos[\theta])\sqrt{\left(\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]\right)\left(1+\frac{(t-tP)^2\ Sin[\theta]^2\ Sin[\phi]^2}{(dG+(t-tP)\ Cos[\theta])^2}\right)\right)}\right) & True \end{cases}$$

$$\begin{cases} -\left(SinSimp[(-t+tP)\ Cos[\phi]\ Sin[\theta],\ dG+(t-tP)\ Cos[\theta]]\middle/\left(2\sqrt{\left(\left(dG^2+(t-tP)^2+\right.\right.}\right.\right. & \left(dG+t\geq tP\ ||\ \theta\leq ArcCos\left[\frac{dG}{-t+tP}\right]\ ||\ \theta+ArcCos\left[\frac{dG}{-t+tP}\right]>2\pi\right)\ \&\&\ (dG+t<tP\ ||\ \theta\leq 0) \end{cases}$$

```
            2 dG (t - tP) Cos[θ])
         (1 + (t-tP)^2 Sin[θ]^2 Sin[ϕ]^2 / (dG+(t-tP) Cos[θ])^2))))
SinSimp[(-t + tP) Cos[ϕ] Sin[θ],                True
  dG + (t - tP) Cos[θ]] /
 (2 √((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[
            θ]) (1 + (t-tP)^2 Sin[θ]^2 Sin[ϕ]^2 / (dG+(t-tP) Cos[θ])^2)))

];
```

In[ ]:= `MatrixForm[h3DIso]`

Out[ ]//MatrixForm=

$$\frac{(dG+(t-tP)\,Cos[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]}\,\left((dG+(t-tP)\,Cos[\theta])^2+(t-tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)}$$

$$\begin{cases} \left((t-tP)^2\,Cos[\phi]\,Sin[\theta]^2\,Sin[\phi]\right)\Big/\left(\left((dG+(t-tP)\,Cos[\theta])^2+(t-tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)\left(1+\frac{(t-tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2}\right)\right)}\right) & \left(dG+t\geq tP \,||\, \theta\leq ArcCos\left[\frac{dG}{-t+tP}\right] \,||\, \theta+ArcCos\left[\frac{dG}{-t+tP}\right]>2\,\pi\right)\,\&\&\,(dG+t<tP \,||\, \theta\leq 0) \\ -\left(\left((t-tP)^2\,Cos[\phi]\,Sin[\theta]^2\,Sin[\phi]\right)\Big/\left(\left((dG+(t-tP)\,Cos[\theta])^2+(t-tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)\left(1+\frac{(t-tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2}\right)\right)}\right)\right) & True \end{cases}$$

$$\begin{cases} \left((t-tP)\,(dG+(t-tP)\,Cos[\theta])\,Cos[\phi]\,Sin[\theta]\right)\Big/\left(\left((dG+(t-tP)\,Cos[\theta])^2+(t-tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)\Big/\sqrt{\left(\left(dG^2\right.\right.}\right. & \left(dG+t\geq tP \,||\, \theta\leq ArcCos\left[\frac{dG}{-t+tP}\right] \,||\, \theta+ArcCos\left[\frac{dG}{-t+tP}\right]>2\,\pi\right)\,\&\&\,(dG+t<tP \,||\, \theta\leq 0) \end{cases}$$

$\text{Cos}[\phi]^2\ \text{Sin}[\theta]^2)\ \sqrt{((\text{dG}^2 +}$
$(\text{t} - \text{tP})^2 + 2\,\text{dG}\,(\text{t} - \text{tP})\,\text{Cos}[\theta])$
$\left(1 + \frac{(\text{t}-\text{tP})^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(\text{dG}+(\text{t}-\text{tP})\,\text{Cos}[\theta])^2}\right)))$

$((-\text{t} + \text{tP})\,(\text{dG} + (\text{t} - \text{tP})\,\text{Cos}[\theta])$ True
$\text{Cos}[\phi]\,\text{Sin}[\theta])\,/$
$(((\text{dG} + (\text{t} - \text{tP})\,\text{Cos}[\theta])^2 + (\text{t} - \text{tP})^2$
$\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2)\ \sqrt{((\text{dG}^2 +}$
$(\text{t} - \text{tP})^2 + 2\,\text{dG}\,(\text{t} - \text{tP})\,\text{Cos}[\theta])$
$\left(1 + \frac{(\text{t}-\text{tP})^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(\text{dG}+(\text{t}-\text{tP})\,\text{Cos}[\theta])^2}\right)))$

Considering If statements

In all cases, the different between the two possible IF statements is a negative sign.

$$\left(\text{dG} + \text{t} \geq \text{tP}\ ||\ \theta \leq \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right]\ ||\ \theta + \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right] > 2\pi\right)\ \&\&\ (\text{dG} + \text{t} < \text{tP}\ ||\ \theta \leq 0)$$

$$\left(\text{dG} + \text{t} \geq \text{tP}\ ||\ \theta \leq \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right]\ ||\ \theta + \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right] > 2\pi\right)\ \&\&\ (\text{dG} + \text{t} < \text{tP})$$

θ is always greater than 0 and is ill - defined when it equals 0.

$$\left(\theta \leq \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right]\ ||\ \theta + \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right] > 2\pi\right)\ \&\&\ (\text{dG} + \text{t} < \text{tP})$$

If $\text{dG} + \text{t} < \text{tP}$ then $\text{dG} + \text{t} \geq \text{tP}$ must be false. $\text{tP} - \text{t} = \text{dP}$ (distance from Earth to the photon), so this is saying $\text{dG} < \text{dP}$.

$$\left(\theta \leq \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right]\right)\ \&\&\ (\text{dG} + \text{t} < \text{tP})$$

The max value of θ is π and if $\text{dP} > \text{dG}$,
$\text{ArcCos}\left[\frac{\text{dG}}{\text{dP}}\right] < \pi$, so $\theta + \text{ArcCos}\left[\frac{\text{dG}}{-\text{t} + \text{tP}}\right] < 2\pi$.

What this If statement is saying is that if $\text{dP} > \text{dG}$ (the distance from Earth to the photon is larger than the distance to the GW source), there is a cuttoff angle given by $\text{ArcCos}\left[\frac{\text{dG}}{\text{dP}}\right]$ below which there is a sign flip for some entries of h_{ij}.

In[ ]:= h3DIso =

$$\frac{(dG+(t-tP)\ \text{Cos}[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \text{Cos}[\theta]}\ \left((dG+(t-tP)\ \text{Cos}[\theta])^2+(t-tP)^2\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\right)}$$

$$\left((t-tP)^2\ \text{Cos}[\phi]\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]\right) \Big/ \left(\left((dG+(t-tP)\ \text{Cos}[\theta])^2+(t-tP)^2\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\right)\sqrt{\left(\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \text{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2}{(dG+(t-tP)\ \text{Cos}[\theta])^2}\right)\right)}\right) \qquad \left(\theta \le \text{ArcCos}\left[\frac{dG}{-t+tP}\right]\right)\ \&\&\ (dG +$$

$$-\left(\left((t-tP)^2\ \text{Cos}[\phi]\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]\right) \Big/ \left(\left((dG+(t-tP)\ \text{Cos}[\theta])^2+(t-tP)^2\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\right)\sqrt{\left(\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \text{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2}{(dG+(t-tP)\ \text{Cos}[\theta])^2}\right)\right)}\right)\right) \qquad \text{True}$$

$$\left((t-tP)\ (dG+(t-tP)\ \text{Cos}[\theta])\ \text{Cos}[\phi]\ \text{Sin}[\theta]\right) \Big/ \left(\left((dG+(t-tP)\ \text{Cos}[\theta])^2+(t-tP)^2\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\right)\sqrt{\left(\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \text{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2}{(dG+(t-tP)\ \text{Cos}[\theta])^2}\right)\right)}\right) \qquad \left(\theta \le \text{ArcCos}\left[\frac{dG}{-t+tP}\right]\right)\ \&\&\ (dG +$$

$$\left((-t+tP)\ (dG+(t-tP)\ \text{Cos}[\theta])\ \text{Cos}[\phi]\ \text{Sin}[\theta]\right) \Big/ \left(\left((dG+(t-tP)\ \text{Cos}[\theta])^2+(t-tP)^2\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\right)\sqrt{\left(\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \text{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2}{(dG+(t-tP)\ \text{Cos}[\theta])^2}\right)\right)}\right) \qquad \text{True}$$

Redshift

In[∘]:=
```
v[θ, ϕ] (*3D looking vector *)
```
Out[∘]=
```
{Cos[ϕ] Sin[θ], Sin[θ] Sin[ϕ], Cos[θ]}
```

In[∘]:=
```
Simplify[v[θ, ϕ].Transpose[v[θ, ϕ].h3DIso]]
```

In[∘]:=

$$
\texttt{H3DIso[t\_] := Cos[}\theta\texttt{] Cos[}\phi\texttt{]}\left(\begin{cases}
((t-tP)\ (dG+(t-tP)\ \mathrm{Cos}[\theta])\ \mathrm{Cos}[\phi]\ \mathrm{Sin}[\theta]) \Big/ \\
\quad \Big(\big((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\big) \\
\qquad \sqrt{\Big(\big(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\big)} \\
\qquad\quad \Big(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\Big)\Big)\Big) \\
((-t+tP)\ (dG+(t-tP)\ \mathrm{Cos}[\theta])\ \mathrm{Cos}[\phi]\ \mathrm{Sin}[\theta]) \Big/ \\
\quad \Big(\big((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\big) \\
\qquad \sqrt{\Big(\big(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\big)} \\
\qquad\quad \Big(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\Big)\Big)\Big)
\end{cases}\right.
$$

$$
\mathrm{Cos}[\phi]\left(\begin{cases}
\big((t-tP)^2\ \mathrm{Cos}[\phi]\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]\big) \Big/ & \theta \le \mathrm{ArcCos}\left[\frac{dG}{-t+tP}\right]\ \&\& \\
\quad \Big(\big((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\big) & \\
\qquad \sqrt{\Big(\big(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\big)} & \\
\qquad\quad \Big(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\Big)\Big)\Big) & \\
-\Big(\big((t-tP)^2\ \mathrm{Cos}[\phi]\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]\big) \Big/ & \mathrm{True} \\
\quad \Big(\big((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\big) & \\
\qquad \sqrt{\Big(\big(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\big)} & \\
\qquad\quad \Big(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\Big)\Big)\Big)\Big) &
\end{cases}\right.
$$

$$
\big((dG+(t-tP)\ \mathrm{Cos}[\theta])^2\ \mathrm{Cos}[\phi]\ \mathrm{Sin}[\theta]\big) \Big/ \Big(\sqrt{dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]}
$$

$$\left((dG + (t - tP)\,Cos[\theta])^2 + (t - tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right) + \left(\left[\begin{array}{l} ((t - tP)^2\,Cos[\phi]\,Sin[\theta]^2\,S \\ \quad (((dG + (t - tP)\,Cos[\theta])^2 \\ \qquad \sqrt{((dG^2 + (t - tP)^2 + 2\,d} \\ \qquad\quad (1 + \frac{(t-tP)^2\,Sin[\theta]^2\,Sin[}{(dG+(t-tP)\,Cos[\theta]} \\ - (((t - tP)^2\,Cos[\phi]\,Sin[\theta] \\ \quad (((dG + (t - tP)\,Cos[\theta] \\ \qquad \sqrt{((dG^2 + (t - tP)^2 + 2} \\ \qquad\quad (1 + \frac{(t-tP)^2\,Sin[\theta]^2\,S}{(dG+(t-tP)\,Cos} \end{array}\right.\right.$$

$$\frac{\begin{array}{l}\Big(2\,(t - tP)\,Cos[\theta]\,(dG + (t - tP)\,Cos[\theta])\,Sin[\theta]^2 \\ \quad \left((dG + (t - tP)\,Cos[\theta])^2 + 2\,(t - tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)\,Sin[\phi]^2\Big) \Big/ \\ \Big(\sqrt{dG^2 + (t - tP)^2 + 2\,dG\,(t - tP)\,Cos[\theta]}\;\Big((dG + (t - tP)\,Cos[\theta])^2 + \\ \quad (t - tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\Big)\,\left((dG + (t - tP)\,Cos[\theta])^2 + (t - tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)\Big) + \\ \quad Sin[\theta]^2\,Sin[\phi]^2\left(-(dG + (t - tP)\,Cos[\theta])^2 + \frac{(t-tP)^4\,Cos[\phi]^2\,Sin[\theta]^4\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2+(t-tP)^2\,Cos[\phi]^2\,Sin[\theta]^2}\right)\end{array}}{\sqrt{dG^2 + (t - tP)^2 + 2\,dG\,(t - tP)\,Cos[\theta]}\;\left((dG + (t - tP)\,Cos[\theta])^2 + (t - tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)}$$

$$\begin{array}{l} + \Big(Cos[\theta]^2\,\Big((t - tP)^2\,(dG + (t - tP)\,Cos[\theta])^2\,Cos[2\,\phi]\,Sin[\theta]^2 - \\ \quad \frac{1}{4}\,(t - tP)^4\,Sin[\theta]^4\,Sin[2\,\phi]^2\Big)\Big) \Big/ \Big(\sqrt{dG^2 + (t - tP)^2 + 2\,dG\,(t - tP)\,Cos[\theta]} \\ \quad \left((dG + (t - tP)\,Cos[\theta])^2 + (t - tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right) \\ \quad \left((dG + (t - tP)\,Cos[\theta])^2 + (t - tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)\Big)\end{array}$$

In[∘]:= ```
ξ3DIso = Simplify[H3DIso[Tmin[tG, tP, dG, θ]]]
```

In[∘]:= ```
χ[dP_, dG_, θ_, ϕ_] :=
  If[θ ≤ M[dP, dG], ArcTan[dP Sin[θ] / (dG - dP Cos[θ])] + π, If[θ ≤ 2 π - M[dP, dG],
    ArcTan[dP Sin[θ] / (dG - dP Cos[θ])], ArcTan[dP Sin[θ] / (dG - dP Cos[θ])] - π]] (*angle between n and R*)
```

In[∘]:= ```
vdotR3D = FullSimplify[Cos[π - θ - χ[DP[tP, Tmin[tG, tP, dG, θ]], dG, θ, ϕ]]]
```

In[∘]:= ```
Simplify[-1/2 1/Abs[1 + vdotR3D] ξ3DIso] (*full redshift equation*)
```

In[∘]:= ```
z3DIsoTrigIfSimp[θ_, ϕ_, dG_, tP_] :=
```

$$\left(\left(\left(\left(\left[\begin{array}{l}(2\,tP\,(2\,dG + tP) \\ \quad Abs\left[2\,dG\,(dG + tP)^2 - \left(4\,dG^3 + 6\,dG^2\,tP + \right.\right. \\ \qquad \left.3\,dG\,tP^2 + tP^3\right)\,Cos[\theta] + \\ \qquad dG\,\left(2\,dG^2 + 2\,dG\,tP + tP^2\right)\,Cos[\theta]^2\Big] \\ \quad \left(-2\,dG\,(dG + tP) + \left(2\,dG^2 + 2\,dG\,tP + tP^2\right)\right. \\ \qquad \left.Cos[\theta]\right)\,Cos[\phi]\,Sin[\theta]\Big) \Big/ \\ \quad \left(\left(2\,dG^2 + 2\,dG\,tP + tP^2 - \right.\right.\end{array}\right.\right.\right.\right.\right. \qquad \begin{array}{l}\theta \le ArcCos\left[-\frac{2\,d}{}\right. \\ 2\,dG^2 < tP^2 + 2\end{array}$$
```

$$\Big(2\ (dG + tP - dG\, Cos[\theta])\ \Big(-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\Big)^2$$
$$Cos[\phi]\ Sin[\theta]\Big) \Big/ \Big(\big(2\, dG^2 + 2\, dG\, tP + tP^2 - 2\, dG\ (dG + tP)\ Cos[\theta]\big)$$
$$\Big(4\, dG^2\ (dG + tP)^2 - 4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP + 3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] +$$
$$\big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\, dG + tP)^2\ Cos[\phi]^2\ Sin[\theta]^2\Big)\Big) +$$

$$
\begin{cases}
\begin{array}{l}
\big(\mathrm{tP}^2\,(2\,\mathrm{dG}+\mathrm{tP})^2 \\
\quad \mathrm{Abs}\big[2\,\mathrm{dG}\,(\mathrm{dG}+\mathrm{tP})^2-\big(4\,\mathrm{dG}^3+6\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+\mathrm{dG}\,\big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)\,\mathrm{Cos}[\theta]^2\big] \\
\quad \mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]\big)\Big/ \\
\Big(\big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2-2\,\mathrm{dG}\,(\mathrm{dG}+\mathrm{tP})\,\mathrm{Cos}[\theta]\big) \\
\quad \big(4\,\mathrm{dG}^2\,(\mathrm{dG}+\mathrm{tP})^2-4\,\mathrm{dG}\,\big(2\,\mathrm{dG}^3+4\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+ \\
\quad \big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)^2\,\mathrm{Cos}[\theta]^2+\mathrm{tP}^2\,(2\,\mathrm{dG}+\mathrm{tP})^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\big) \\
\quad \sqrt{\big(4\,\mathrm{dG}^2\,(\mathrm{dG}+\mathrm{tP})^2-4\,\mathrm{dG}\,\big(2\,\mathrm{dG}^3+4\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+ \big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)^2\,\mathrm{Cos}[\theta]^2+\mathrm{tP}^2\,(2\,\mathrm{dG}+\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\big)}\Big)
\end{array}
& \begin{array}{l}\theta \le \mathrm{ArcCos}\Big[-\frac{2\,\mathrm{dG}\,(-\mathrm{dG}-\mathrm{tP}+\mathrm{dG}\,\mathrm{Cos}[\theta])}{\mathrm{tP}\,(2\,\mathrm{dG}+\mathrm{tP})} \\ 2\,\mathrm{dG}^2<\mathrm{tP}^2+2\,\mathrm{dG}^2\,\mathrm{Cos}[\theta]\end{array} \\
\begin{array}{l}
-\Big(\big(\mathrm{tP}^2\,(2\,\mathrm{dG}+\mathrm{tP})^2 \\
\quad \mathrm{Abs}\big[2\,\mathrm{dG}\,(\mathrm{dG}+\mathrm{tP})^2-\big(4\,\mathrm{dG}^3+6\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+\mathrm{dG}\,\big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)\,\mathrm{Cos}[\theta]^2\big] \\
\quad \mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]\big)\Big/ \\
\Big(\big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2-2\,\mathrm{dG}\,(\mathrm{dG}+\mathrm{tP})\,\mathrm{Cos}[\theta]\big) \\
\quad \big(4\,\mathrm{dG}^2\,(\mathrm{dG}+\mathrm{tP})^2-4\,\mathrm{dG}\,\big(2\,\mathrm{dG}^3+4\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+ \\
\quad \big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)^2\,\mathrm{Cos}[\theta]^2+\mathrm{tP}^2\,(2\,\mathrm{dG}+\mathrm{tP})^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\big) \\
\quad \sqrt{\big(4\,\mathrm{dG}^2\,(\mathrm{dG}+\mathrm{tP})^2-4\,\mathrm{dG}\,\big(2\,\mathrm{dG}^3+4\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+ \big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)^2\,\mathrm{Cos}[\theta]^2+\mathrm{tP}^2\,(2\,\mathrm{dG}+\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\big)}\Big)\Big)
\end{array}
& \mathrm{True}
\end{cases}
$$

$$
(\mathrm{dG}+\mathrm{tP}-\mathrm{dG}\,\mathrm{Cos}[\theta])\,\big(-2\,\mathrm{dG}\,(\mathrm{dG}+\mathrm{tP})+\big(2\,\mathrm{dG}^2+2\,\mathrm{dG}\,\mathrm{tP}+\mathrm{tP}^2\big)\,\mathrm{Cos}[\theta]\big)
$$
$$
\mathrm{Sin}[\theta]\,\big(4\,\mathrm{dG}^2\,(\mathrm{dG}+\mathrm{tP})^2-4\,\mathrm{dG}\,\big(2\,\mathrm{dG}^3+4\,\mathrm{dG}^2\,\mathrm{tP}+3\,\mathrm{dG}\,\mathrm{tP}^2+\mathrm{tP}^3\big)\,\mathrm{Cos}[\theta]+
$$

$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 \mathrm{Cos}[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\Big)$$

$$\mathrm{Sin}[\phi]^2\Big)\Big/\Big(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\right)$$

$$\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$

$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 \mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\Big)$$

$$\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$

$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 \mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)\Big)+$$

$$\Bigg(4\,(dG+tP-dG\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]\,\mathrm{Sin}[\phi]^2\left(-\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^2+\right.$$

$$\left(tP^4\,(2\,dG+tP)^4\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^4\,\mathrm{Sin}[\phi]^2\right)\Big/\Big(4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2$$

$$\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$

$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 \mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\Big)\Big)\right)\Bigg)\Big/$$

$$\Bigg(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\right)\left(\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^2+\right.$$

$$\left.\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\right)\Bigg)+$$

$$\left(\begin{cases} \left(tP^2\ (2\ dG + tP)^2\ \mathrm{Abs}\left[2\ dG\ (dG + tP)^2 - \left(4\ dG^3 + 6\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + dG\ \left(2\ dG^2 + 2\ dG\ tP + tP^2\right) \mathrm{Cos}[\theta]^2\right] \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[2\ \phi]\right) \Big/ \\ \quad \Big(\left(2\ dG^2 + 2\ dG\ tP + tP^2 - 2\ dG\ (dG + tP)\ \mathrm{Cos}[\theta]\right) \\ \quad \left(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \left(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 + tP^2\ (2\ dG + tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\right) \\ \quad \sqrt{\left(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \left(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 + tP^2\ (2\ dG + tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2\right)}\Big) & \theta \le \mathrm{ArcCos}\left[-\frac{2\ dG\ (-dG - tP + dG\ \mathrm{Cos}[\theta])}{tP\ (2\ dG + tP)}\right] \ \&\& \ 2\ dG^2 < tP^2 + 2\ dG^2\ \mathrm{Cos}[\theta] \\ -\Big(\left(tP^2\ (2\ dG + tP)^2\ \mathrm{Abs}\left[2\ dG\ (dG + tP)^2 - \left(4\ dG^3 + 6\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + dG\ \left(2\ dG^2 + 2\ dG\ tP + tP^2\right) \mathrm{Cos}[\theta]^2\right] \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[2\ \phi]\right) \Big/ \\ \quad \Big(\left(2\ dG^2 + 2\ dG\ tP + tP^2 - 2\ dG\ (dG + tP)\ \mathrm{Cos}[\theta]\right) \\ \quad \left(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \left(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 + tP^2\ (2\ dG + tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\right) \\ \quad \sqrt{\left(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \left(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 + tP^2\ (2\ dG + tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2\right)}\Big)\Big) & \mathrm{True} \end{cases}\right)$$

$$\mathrm{Sin}[\theta]\ \left(4\ \left(-2\ dG\ (dG + tP) + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right) \mathrm{Cos}[\theta]\right)^2 \mathrm{Cos}[2\ \phi] - tP^2\ (2\ dG + tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[2\ \phi]^2\right) \Big/$$

$$\Big(\left(2\ dG^2 + 2\ dG\ tP + tP^2 - 2\ dG\ (dG + tP)\ \mathrm{Cos}[\theta]\right)\ \Big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \left(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right) \mathrm{Cos}[\theta] + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 +$$

$$tP^2\ (2\ dG + tP)^2\ Cos[\phi]^2\ Sin[\theta]^2\Big)\ \Big(4\ dG^2\ (dG + tP)^2 -$$

$$4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 +$$

$$tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\Big)\Big)\Bigg)\Bigg) \Big/ \Bigg(4\ Abs\Big[1 + \Bigg(\Big[-Cos\Big[\theta - ArcTan\Big[\frac{tP\ (2\ dG+}{-2\ dG\ (dG+tP) + (2\ dG}$$

$$Cos\Big[\theta - ArcTan\Big[\frac{tP\ (2\ dG}{-2\ dG\ (dG+tP) + (2\ d}$$

```
z3DIsoTrigSimp[θ_, ϕ_, dG_, tP_] :=
 - ((Sin[θ] (2 Cos[θ] Cos[ϕ] (
   (2 tP (2 dG + tP)                                              (2 dG² ≥ tP² + 2
      Abs[2 dG (dG + tP)² - (4 dG³ + 6 dG² tP +                      θ ≤ ArcCos[
           3 dG tP² + tP³) Cos[θ] +                                  θ + ArcCos[
        dG (2 dG² + 2 dG tP + tP²) Cos[θ]²]                          (2 dG² < tP² + 2
      (-2 dG (dG + tP) + (2 dG² + 2 dG tP + tP²)
         Cos[θ]) Cos[ϕ] Sin[θ]) /
    ((2 dG² + 2 dG tP + tP² -
        2 dG (dG + tP) Cos[θ])
      (4 dG² (dG + tP)² - 4 dG (2 dG³ +
           4 dG² tP + 3 dG tP² + tP³) Cos[θ] +
        (2 dG² + 2 dG tP + tP²)² Cos[θ]² +
        tP² (2 dG + tP)² Cos[ϕ]² Sin[θ]²)
      √(4 dG² (dG + tP)² - 4 dG (2 dG³ +
             4 dG² tP + 3 dG tP² + tP³) Cos[θ] +
          (2 dG² + 2 dG tP + tP²)² Cos[θ]² +
          tP² (2 dG + tP)² Sin[θ]² Sin[ϕ]²))
   -((2 tP (2 dG + tP)                                              True
        Abs[2 dG (dG + tP)² - (4 dG³ + 6 dG² tP +
             3 dG tP² + tP³) Cos[θ] +
          dG (2 dG² + 2 dG tP + tP²) Cos[θ]²]
        (-2 dG (dG + tP) +
          (2 dG² + 2 dG tP + tP²) Cos[θ])
        Cos[ϕ] Sin[θ]) /
      ((2 dG² + 2 dG tP + tP² -
          2 dG (dG + tP) Cos[θ])
        (4 dG² (dG + tP)² - 4 dG (2 dG³ +
             4 dG² tP + 3 dG tP² + tP³) Cos[θ] +
          (2 dG² + 2 dG tP + tP²)² Cos[θ]² +
          tP² (2 dG + tP)² Cos[ϕ]² Sin[θ]²)
        √(4 dG² (dG + tP)² - 4 dG (2 dG³ + 4 dG²
               tP + 3 dG tP² + tP³) Cos[θ] +
            (2 dG² + 2 dG tP + tP²)² Cos[θ]² +
            tP² (2 dG + tP)² Sin[θ]² Sin[ϕ]²)))
```

$$\left(2\ (dG + tP - dG\ \mathrm{Cos}[\theta])\ \left(-2\ dG\ (dG + tP) + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)\ \mathrm{Cos}[\theta]\right)^2 \mathrm{Cos}[\phi]\ \mathrm{Sin}[\theta]\right) \Big/ \left(\left(2\ dG^2 + 2\ dG\ tP + tP^2 - 2\ dG\ (dG + tP)\ \mathrm{Cos}[\theta]\right) \left(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \left(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\right)\ \mathrm{Cos}[\theta] + \left(2\ dG^2 + 2\ dG\ tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 + tP^2\ (2\ dG + tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\right)\right) +$$

$$\begin{cases} \left(tP^2\,(2\,dG+tP)^2\,\text{Abs}\left[2\,dG\,(dG+tP)^2-\left(4\,dG^3+6\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+dG\left(2\,dG^2+2\,dG\,tP+tP^2\right)\text{Cos}[\theta]^2\right]\text{Sin}[\theta]^2\,\text{Sin}[2\,\phi]\right)\Big/\left(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\text{Cos}[\theta]\right)\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\right)\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\right)}\right) & \left(2\,dG^2\ge tP^2+2\,dG^2\,\text{Cos}[\theta]\;||\;\theta\le\text{ArcCos}\left[-\frac{2\,dG\,(-dG-tP+dG\,\text{Cos}}{tP\,(2\,dG+tP)}\right.\right.\;\theta+\text{ArcCos}\left[-\frac{2\,dG\,(-dG-tP+dG\,\text{Cos}[}{tP\,(2\,dG+tP)}\right.\;\left(2\,dG^2<tP^2+2\,dG^2\,\text{Cos}[\theta]\;||\;\theta\right. \\ -\left(\left(tP^2\,(2\,dG+tP)^2\,\text{Abs}\left[2\,dG\,(dG+tP)^2-\left(4\,dG^3+6\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+dG\left(2\,dG^2+2\,dG\,tP+tP^2\right)\text{Cos}[\theta]^2\right]\text{Sin}[\theta]^2\,\text{Sin}[2\,\phi]\right)\Big/\left(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\text{Cos}[\theta]\right)\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\right)\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\right)}\right)\right) & \text{True} \end{cases}$$

$$(dG+tP-dG\,\text{Cos}[\theta])\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\text{Cos}[\theta]\right)$$
$$\text{Sin}[\theta]\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\text{Cos}[\theta]+\right.$$

$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\Big)$$

$$Sin[\phi]^2\Big)\Big/\left(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,Cos[\theta]\right)\right.$$

$$\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\right.$$

$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)$$

$$\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\right.$$

$$\left.\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)\right)+$$

$$\left(4\,(dG+tP-dG\,Cos[\theta])\,Sin[\theta]\,Sin[\phi]^2\left(-\left(dG+\frac{tP\,(2\,dG+tP)\,Cos[\theta]}{-2\,(dG+tP)+2\,dG\,Cos[\theta]}\right)^2+\right.\right.$$

$$\left(tP^4\,(2\,dG+tP)^4\,Cos[\phi]^2\,Sin[\theta]^4\,Sin[\phi]^2\right)\Big/\left(4\,(dG+tP-dG\,Cos[\theta])^2\right.$$

$$\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\right.$$

$$\left.\left.\left.\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)\right)\right)\right)\Big/$$

$$\left(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,Cos[\theta]\right)\left(\left(dG+\frac{tP\,(2\,dG+tP)\,Cos[\theta]}{-2\,(dG+tP)+2\,dG\,Cos[\theta]}\right)^2+\right.\right.$$

$$\left.\left.\frac{tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{4\,(dG+tP-dG\,Cos[\theta])^2}\right)\right)+$$

$$
\begin{cases}
\Big(tP^2 (2\,dG + tP)^2\, \mathrm{Abs}\big[2\,dG\,(dG+tP)^2 - \big(4\,dG^3 + 6\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + dG\,\big(2\,dG^2 + 2\,dG\,tP + tP^2\big)\,\mathrm{Cos}[\theta]^2\big]\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]\Big) \Big/ \\
\quad \Big(\big(2\,dG^2 + 2\,dG\,tP + tP^2 - 2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\big) \\
\quad \big(4\,dG^2 (dG+tP)^2 - 4\,dG\,\big(2\,dG^3 + 4\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + \big(2\,dG^2 + 2\,dG\,tP + tP^2\big)^2\,\mathrm{Cos}[\theta]^2 + tP^2 (2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\big) \\
\quad \sqrt{\big(4\,dG^2 (dG+tP)^2 - 4\,dG\,\big(2\,dG^3 + 4\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + \big(2\,dG^2 + 2\,dG\,tP + tP^2\big)^2\,\mathrm{Cos}[\theta]^2 + tP^2 (2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\big)}\Big) & \big(2\,dG^2 \geq tP^2 + 2\,dG^2\,\mathrm{Cos}[\theta] \,||\, \theta \leq \mathrm{ArcCos}\big[-\frac{2\,dG\,(-dG-tP+dG\,\mathrm{Cos}[\theta])}{tP\,(2\,dG+tP)}\big. \\
 & \theta + \mathrm{ArcCos}\big[-\frac{2\,dG\,(-dG-tP+dG\,\mathrm{Cos}[\theta])}{tP\,(2\,dG+tP)}\big] \\
 & \big(2\,dG^2 < tP^2 + 2\,dG^2\,\mathrm{Cos}[\theta] \,||\, \theta \leq 0 \\
-\Big(\Big(tP^2 (2\,dG + tP)^2\, \mathrm{Abs}\big[2\,dG\,(dG+tP)^2 - \big(4\,dG^3 + 6\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + dG\,\big(2\,dG^2 + 2\,dG\,tP + tP^2\big)\,\mathrm{Cos}[\theta]^2\big]\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]\Big) \Big/ \\
\quad \Big(\big(2\,dG^2 + 2\,dG\,tP + tP^2 - 2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\big) \\
\quad \big(4\,dG^2 (dG+tP)^2 - 4\,dG\,\big(2\,dG^3 + 4\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + \big(2\,dG^2 + 2\,dG\,tP + tP^2\big)^2\,\mathrm{Cos}[\theta]^2 + tP^2 (2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\big) \\
\quad \sqrt{\big(4\,dG^2 (dG+tP)^2 - 4\,dG\,\big(2\,dG^3 + 4\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + \big(2\,dG^2 + 2\,dG\,tP + tP^2\big)^2\,\mathrm{Cos}[\theta]^2 + tP^2 (2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\big)}\Big)\Big) & \mathrm{True}
\end{cases}
$$

$$
\mathrm{Sin}[\theta]\,\Big(4\,\big(-2\,dG\,(dG+tP) + \big(2\,dG^2 + 2\,dG\,tP + tP^2\big)\,\mathrm{Cos}[\theta]\big)^2\,\mathrm{Cos}[2\,\phi] - tP^2 (2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]^2\Big)\Big) \Big/
$$

$$
\Big(\big(2\,dG^2 + 2\,dG\,tP + tP^2 - 2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\big)\,\big(4\,dG^2 (dG+tP)^2 - 4\,dG\,\big(2\,dG^3 + 4\,dG^2\,tP + 3\,dG\,tP^2 + tP^3\big)\,\mathrm{Cos}[\theta] + \big(2\,dG^2 + 2\,dG\,tP + tP^2\big)^2\,\mathrm{Cos}[\theta]^2 +
$$

```
tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2) (4 dG^2 (dG + tP)^2 -
4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] + (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 +

tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)) / (4 Abs[1 + ((-Cos[θ -
        ArcTan[tP (2 dG+ / (-2 dG (dG+tP) + (2 dG
   Cos[
    θ - ArcTan[tP (2 dG / (-2 dG (dG+tP) + (2 d
```

In[ ]:=
```
z3DIso[θ_, ϕ_, dG_, tP_] :=
```



```
(( ( ( [ (Abs[2 dG (dG + tP)^2 -          (2 dG^2 ≥ tP^2 + 2 d
```

$$\text{Cos}[\phi]\ \text{Cos}[\theta]\ \Bigg(\begin{cases} \Big(\text{Abs}\Big[2\,\text{dG}\,(\text{dG}+\text{tP})^2 - (4\,\text{dG}^3+6\,\text{dG}^2\,\text{tP}+3\,\text{dG}\,\text{tP}^2+\text{tP}^3)\,\text{Cos}[\theta]+\text{dG}\,(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)\,\text{Cos}[\theta]^2\Big]\,\text{Sin}\Big[2\,\text{ArcTan}\Big[\frac{\text{tP}\,(2\,\text{dG}+\text{tP})\,\text{Cos}[\phi]\,\text{Sin}[\theta]}{-2\,\text{dG}\,(\text{dG}+\text{tP})+(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)\,\text{Cos}[\theta]}\Big]\Big]\Big)\Big/\Big(\big(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2-2\,\text{dG}\,(\text{dG}+\text{tP})\,\text{Cos}[\theta]\big)\sqrt{\big(4\,\text{dG}^2\,(\text{dG}+\text{tP})^2-4\,\text{dG}\,(2\,\text{dG}^3+4\,\text{dG}^2\,\text{tP}+3\,\text{dG}\,\text{tP}^2+\text{tP}^3)\,\text{Cos}[\theta]+(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)^2\,\text{Cos}[\theta]^2+\text{tP}^2\,(2\,\text{dG}+\text{tP})^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)}\Big) & \Big(2\,\text{dG}^2 \geq \text{tP}^2+2\,\text{dG}^2 \ldots\ \theta \leq \text{ArcCos}\Big[-\frac{2\ldots}{\ldots} \ldots\ \theta+\text{ArcCos}\Big[-\frac{2\,\text{d}\ldots}{\ldots} \ldots\ \big(2\,\text{dG}^2 < \text{tP}^2+2\,\text{dG}\ldots \\ -\Big(\Big(\text{Abs}\Big[2\,\text{dG}\,(\text{dG}+\text{tP})^2 - (4\,\text{dG}^3+6\,\text{dG}^2\,\text{tP}+3\,\text{dG}\,\text{tP}^2+\text{tP}^3)\,\text{Cos}[\theta]+\text{dG}\,(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)\,\text{Cos}[\theta]^2\Big]\,\text{Sin}\Big[2\,\text{ArcTan}\Big[\frac{\text{tP}\,(2\,\text{dG}+\text{tP})\,\text{Cos}[\phi]\,\text{Sin}[\theta]}{-2\,\text{dG}\,(\text{dG}+\text{tP})+(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)\,\text{Cos}[\theta]}\Big]\Big]\Big)\Big/\Big(\big(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2-2\,\text{dG}\,(\text{dG}+\text{tP})\,\text{Cos}[\theta]\big)\sqrt{\big(4\,\text{dG}^2\,(\text{dG}+\text{tP})^2-4\,\text{dG}\,(2\,\text{dG}^3+4\,\text{dG}^2\,\text{tP}+3\,\text{dG}\,\text{tP}^2+\text{tP}^3)\,\text{Cos}[\theta]+(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)^2\,\text{Cos}[\theta]^2+\text{tP}^2\,(2\,\text{dG}+\text{tP})^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)}\Big)\Big) & \text{True} \end{cases}\Bigg)$$

$$\Big(-2\,\text{dG}\,(\text{dG}+\text{tP})+(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)\,\text{Cos}[\theta]\Big)^2\,\text{Cos}[\phi]\,\text{Sin}[\theta]\Big)\Big/$$

$$\Big(\big(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2-2\,\text{dG}\,(\text{dG}+\text{tP})\,\text{Cos}[\theta]\big)\,\Big(4\,\text{dG}^2\,(\text{dG}+\text{tP})^2-4\,\text{dG}\,(2\,\text{dG}^3+4\,\text{dG}^2\,\text{tP}+3\,\text{dG}\,\text{tP}^2+\text{tP}^3)\,\text{Cos}[\theta]+(2\,\text{dG}^2+2\,\text{dG}\,\text{tP}+\text{tP}^2)^2\,\text{Cos}[\theta]^2+$$

(tP (2 dG + tP)
Abs[2 dG (dG + tP)^2 - (4 dG^3 + 6 dG
3 dG tP^2 + tP^3) Cos[θ] +
dG (2 dG^2 + 2 dG tP + tP^2) Cos[θ
Sin[θ] Sin[ϕ] Sin[2 ArcTan[
tP (2 dG+tP) Cos[ϕ] Sin[θ] / -2 dG (dG+tP) + (2 dG^2+2 dG tP+tP^2) Cos[θ
((2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG + tP
Cos[θ]) (-2 dG (dG + tP) +
(2 dG^2 + 2 dG tP + tP^2) Cos[θ])
√(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 +
4 dG^2 tP + 3 dG tP^2 + tP^3) C
(2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ
tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[

tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2)) +

-((tP (2 dG + tP)
Abs[2 dG (dG + tP)^2 - (4 dG^3 + 6
3 dG tP^2 + tP^3) Cos[θ] +
dG (2 dG^2 + 2 dG tP + tP^2) Cos
Sin[θ] Sin[ϕ] Sin[2 ArcTan[
tP (2 dG+tP) Cos[ϕ] Sin[θ] / -2 dG (dG+tP) + (2 dG^2+2 dG tP+tP^2) C
((2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG +
Cos[θ]) (-2 dG (dG + tP) +
(2 dG^2 + 2 dG tP + tP^2) Cos[θ
√(4 dG^2 (dG + tP)^2 - 4 dG (2 dG
dG^2 tP + 3 dG tP^2 + tP^3)
(2 dG^2 + 2 dG tP + tP^2)^2 Cos
tP^2 (2 dG + tP)^2 Sin[θ]^2 Si

(8 (dG + tP - dG Cos[θ])^3 Sin[θ] Sin[ϕ]^2 (-(dG + tP (2 dG + tP) Cos[θ] / (-2 (dG + tP) + 2 dG Cos[θ]))^2 +
(tP^4 (2 dG + tP)^4 Cos[ϕ]^2 Sin[θ]^4 Sin[ϕ]^2) / (4 (dG + tP - dG Cos[θ])^2
(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
(2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2)))) /
((2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG + tP) Cos[θ]) (4 dG^2 (dG + tP)^2 -

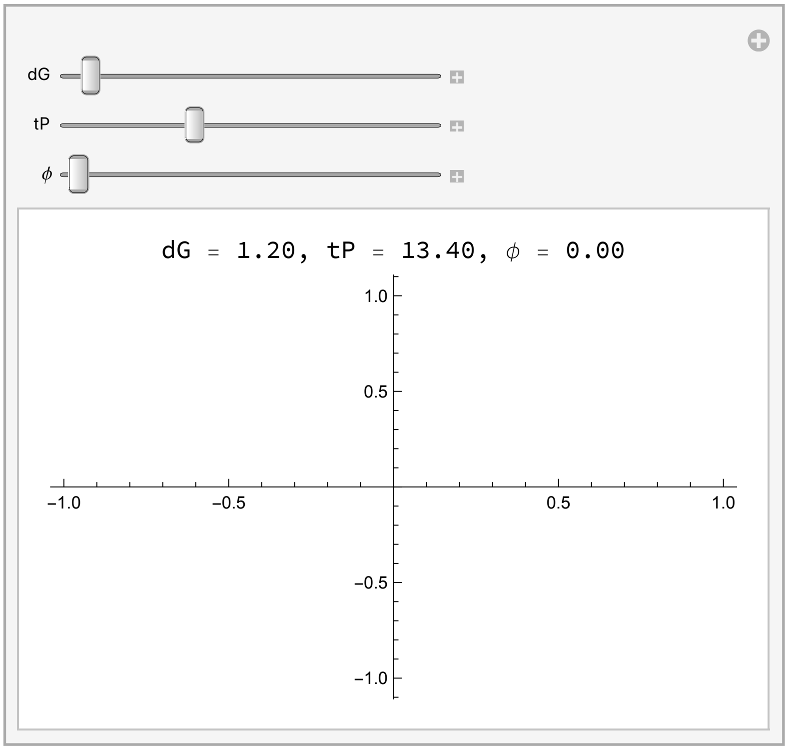
$$4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$
$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)\Big)+$$
$$\Big(2\,tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\theta]^2\,(dG+tP-dG\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]$$
$$\Big(-\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)^2\mathrm{Sin}[\phi]^2+$$
$$\mathrm{Cos}[\phi]^2\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$
$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2-tP^4\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)-dG\,tP^2\,(dG+tP)$$
$$\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]^2\Big)\Big)\Big/\Big(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\right)$$
$$\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$
$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\Big)$$
$$\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$
$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)\Big)+\Big(2\,\mathrm{Cos}[\theta]$$
$$(dG+tP-dG\,\mathrm{Cos}[\theta])\,\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)$$
$$\mathrm{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\Big)$$
$$\mathrm{Sin}[\phi]\,\mathrm{Sin}\Big[2\,\mathrm{ArcTan}\Big[\frac{tP\,(2\,dG+tP)\,\mathrm{Sin}[\theta]\,\mathrm{Sin}[\phi]}{-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]}\Big]\Big]\Big)\Big/$$
$$\Big(\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\right)\Big(4\,dG^2\,(dG+tP)^2-$$
$$4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+$$

$$
\left.\left.\left.\left. tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)\right)\right)\right) \Big/ \left(2\,\mathrm{Abs}\left[1+\left(\left[\begin{array}{l} -\mathrm{Cos}\Big[\theta - \\ \quad \mathrm{ArcTan}\Big[\dfrac{tP\,(2\,dG+}{-2\,dG\,(dG+tP)+(2\,dG} \\ \\ \mathrm{Cos}\Big[\\ \quad \theta-\mathrm{ArcTan}\Big[\dfrac{tP\,(2\,dG}{-2\,dG\,(dG+tP)+(2\,d} \end{array}\right.\right.\right.\right.
$$

Plotting

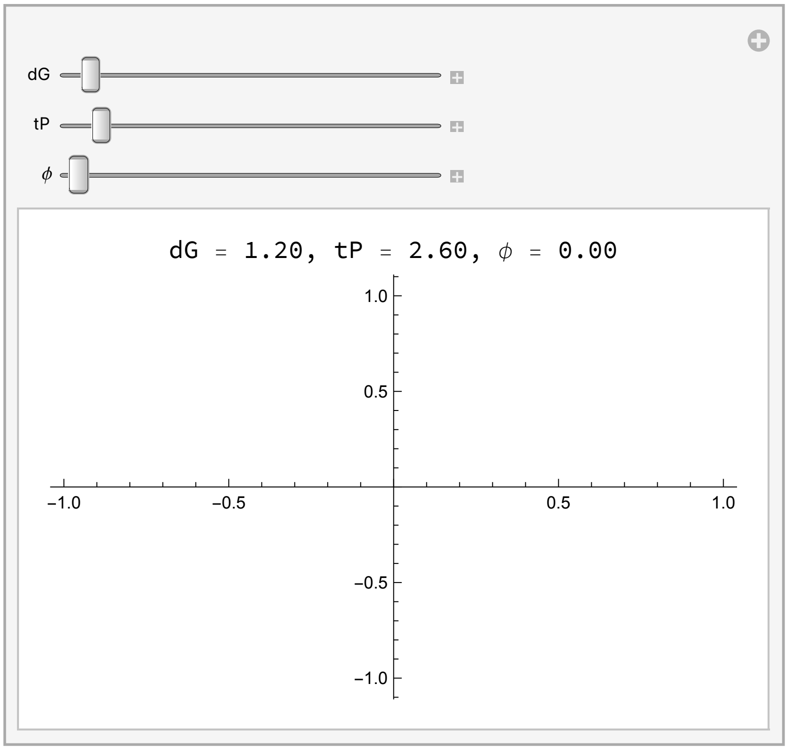
In[ ]:=
```
Manipulate[
 Labeled[Plot[z3DIsoTrigIfSimp[θ, ϕ, dG, tP] , {θ, 0, π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", ϕ = ",
    NumberForm[ϕ, {3, 2}]}], Top], {dG, 0.1, 30}, {tP, 0.01, 40}, {ϕ, 0, 2 π}]
```

Out[ ]=



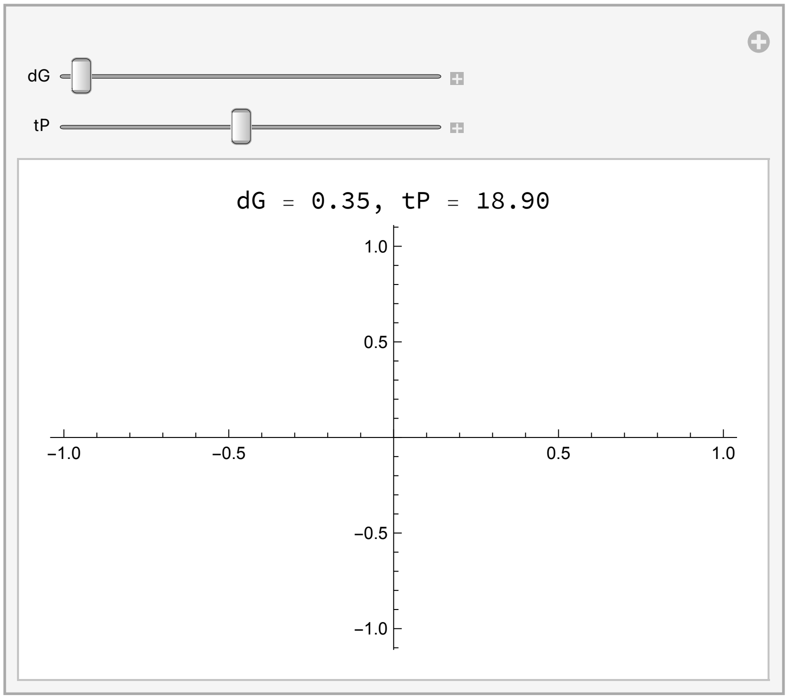
```
Manipulate[Labeled[Plot[z3DIso[θ, ϕ, dG, tP] , {θ, 0, π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", ϕ = ",
    NumberForm[ϕ, {3, 2}]}], Top], {dG, 0.1, 30}, {tP, 0.01, 40}, {ϕ, 0, 2 π}]
```

Out[ ]=



In[ ]:=
```
Manipulate[Labeled[Plot[zSimp[θ, dG, tP] , {θ, 0, π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}] , ", tP = ", NumberForm[tP, {3, 2}]}],
  Top] , {dG, 0.1, 30}, {tP, 0.01, 40}]
```

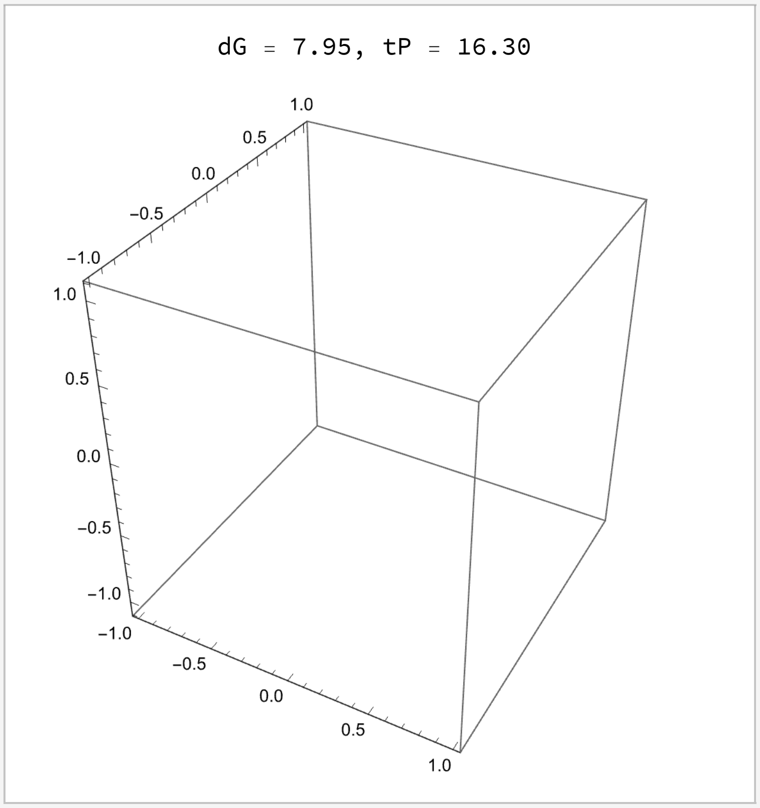
Out[ ]=



In[ ]:=
```
Manipulate[Labeled[SphericalPlot3D[z3DIso[θ, ϕ, dG, tP] ,
   {θ, 0.001, π - 0.001}, {ϕ, 0.001, 2 π - 0.001}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}] , ", tP = ", NumberForm[tP, {3, 2}],}],
  Top] , {dG, 0.1, 30}, {tP, 0.01, 40}]
```

Out[ ]=

dG
tP



Max : Invalid comparison with 6.05793 − 0.611681 i attempted.

Max : Invalid comparison with 6.05793 − 0.611681 i attempted.

Max : Invalid comparison with 0.225257 − 0.611681 i attempted.

General : Further output of Max::nord will be suppressed during this calculation.

Min : Comparison of Max [−6.05793 + 0.611681 i, −0.225257 + 0.611681 i, 0.125263] and ∞ is invalid.

Min : Comparison of Max [−6.05793 + 0.611681 i, −0.225257 + 0.611681 i, 0.000405264] and ∞ is invalid.

Min : Comparison of Max [0.0207383, 6.05793 − 0.611681 i] and ∞ is invalid.

General : Further output of Min::nord2 will be suppressed during this calculation.

Max : Invalid comparison with 6.05793 − 3.74048 i attempted.

Max : Invalid comparison with 6.05793 − 3.74048 i attempted.

Max : Invalid comparison with 0.225257 − 3.74048 i attempted.

General : Further output of Max::nord will be suppressed during this calculation.

Min : Comparison of Max [−6.05793 + 3.74048 i, −0.225257 + 3.74048 i, 3.1933] and ∞ is invalid.

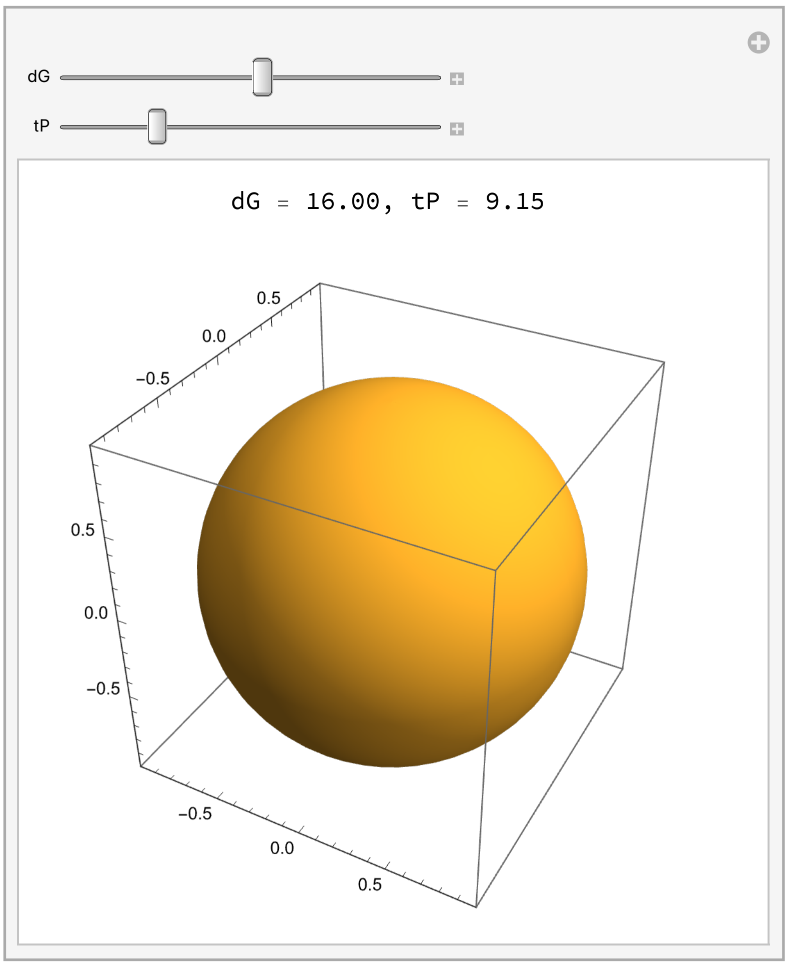
Min : Comparison of Max [3.1933, 6.05793 − 3.74048 i] and ∞ is invalid.

Min : Comparison of Max [0.225257 − 3.74048 i, 3.1933] and ∞ is invalid.

General : Further output of Min::nord2 will be suppressed during this calculation.

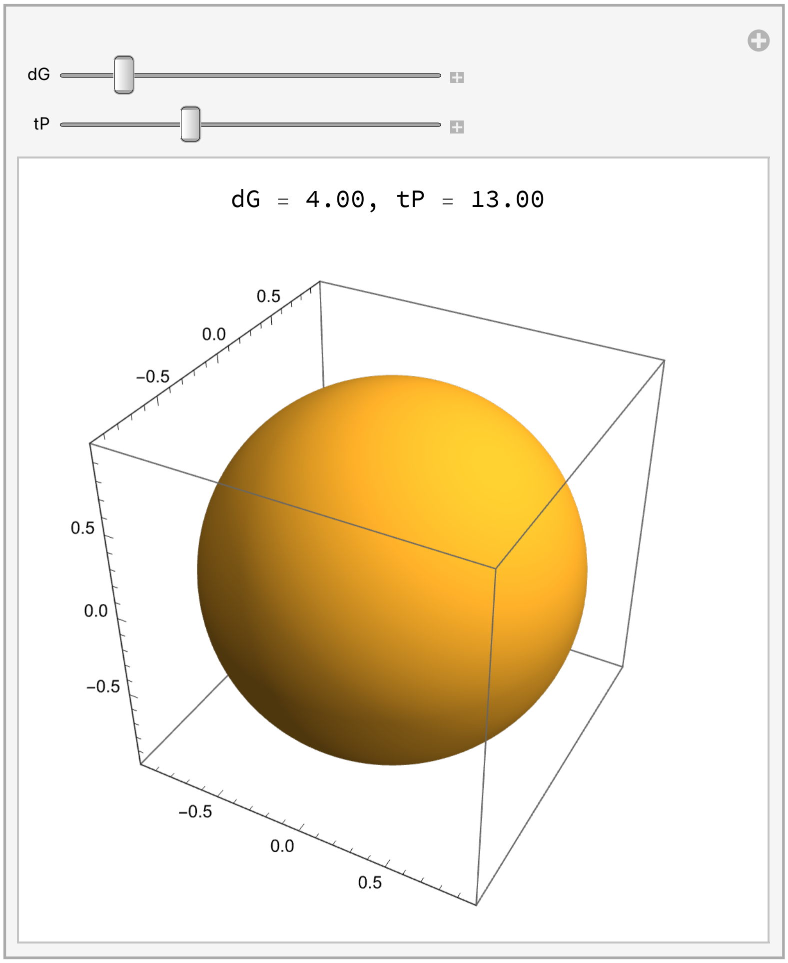
In[ ]:=
```
Manipulate[Labeled[SphericalPlot3D[1, {θ, 0, Pi},
   {φ, 0, 2 Pi}, ColorFunction → Function[{θ, ϕ}, z3DIso[θ, ϕ, dG, tP]],
   ColorFunctionScaling → True, PlotPoints → 50, Lighting → "Neutral", Mesh → None],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],}],
  Top], {dG, 0.1, 30}, {tP, 0.01, 40}]
```

Out[ ]=



In[ ]:=
```
Manipulate[Labeled[SphericalPlot3D[1, {θ, 0, Pi}, {φ, 0, 2 Pi}, ColorFunction →
    Function[{θ, ϕ}, ColorData["TemperatureMap"] × z3DIso[θ, ϕ, dG, tP]],
   ColorFunctionScaling → False, PlotPoints → 100, Lighting → "Neutral", Mesh → None],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],}],
  Top], {dG, 0.1, 30}, {tP, 0.01, 40}]
```

Out[ ]=



3D, Anisotropic, Plus-Polarised Wave: Work in Progress